

00 — Research Map

From Traditional Forecasting to Large Language Models: Evaluating AI-Based Inventory Optimization

Central research question

How does the effectiveness of AI-based inventory optimization change from traditional forecasting models to large language model-based approaches?

This notebook is the *dashboard* for the entire investigation. It is readable without running every experiment: it records the question, hypotheses, model ladder, datasets, evaluation framework, key decisions, and current progress.

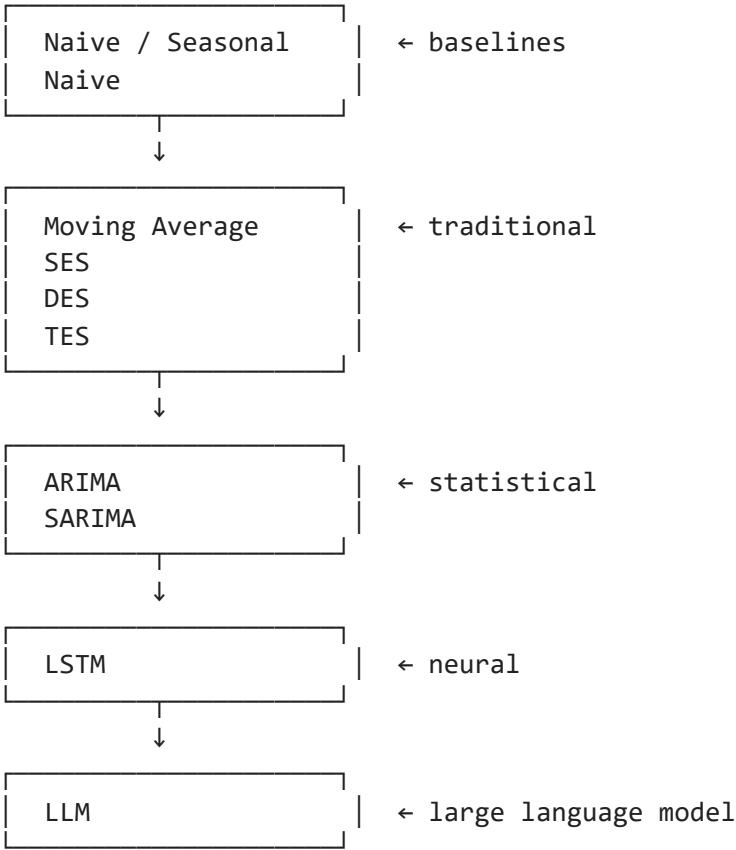
Hypotheses (neutral framing)

#	Statement	Interpretation if rejected
H1	Forecast accuracy differs significantly across model generations on the same series and horizons.	Simple methods already capture the signal; added complexity doesn't improve accuracy.
H2	Forecast-accuracy differences translate into measurable inventory-outcome differences under a common policy.	Inventory outcomes are robust to forecast quality; policy design matters more than forecast precision.
H3	Increasing model sophistication yields <i>monotonically</i> improving inventory outcomes.	A simpler model may outperform a complex one in inventory terms — a publishable finding.
H4	Observed patterns are consistent across the two retail environments (M5 and Store Item Demand).	The findings are dataset-specific and may not generalise.

No hypothesis is privileged. A negative finding is as valuable as a positive one.

The model ladder

The models progress from traditional smoothing → statistical → neural → LLM-based approaches. Complexity increases along the ladder; whether *value* increases with complexity is the empirical question this study answers.



Every model feeds the **same** inventory simulation — differences in outcomes are attributable to differences in forecasts, not in policy design.

Datasets

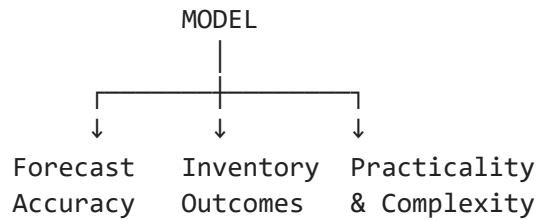
Dataset	Role	Source	What it allows
Walmart M5	Primary	Kaggle M5 Forecasting — Accuracy	30,490 item-store series; daily; ~5 years; hierarchical retail demand; established benchmark
Store Item Demand Forecasting	Robustness (RQ4/H4)	Kaggle Store Item Demand Challenge	10 stores x 50 items, daily, complete panel

Status (2026-08-27): Store Item Demand Forecasting is now Dataset 02 (500 series, 1826 days) available (2013-01-01 to 2013-07-17) with a corrupted archive. The full dataset would cover ~4.5 years. This limits cross-dataset robustness analysis to a **planned** but **currently infeasible** second environment.

Common calendar window (user constraint, 2026-08-27): Both datasets must share the same primary date window with equal observations per experimental series, established from the actual temporal overlap. See `02_data_acquisition_and_audit.ipynb` for the evidence.

Evaluation framework

Three distinct dimensions, none reducible to the others:



Forecast accuracy — MAE, RMSE, RMSSE (scale-independent), sMAPE, MASE.

Inventory outcomes — total cost (holding + stockout), stockout rate, service level, average inventory, order frequency. A common order-up-to (R, S) policy with identical parameters for every model.

Practicality — runtime, RAM, complexity, interpretability, reproducibility, latency.

Critical insight: better forecast accuracy does not automatically yield better inventory outcomes. A model with *worse* MAE can produce *better* inventory cost if its error structure is less harmful. This is one of the most important business findings the study may produce.

Key design principles

Principle	What it means in practice
Neutrality	The LLM is not expected to win. Any outcome is informative.
Time integrity	No future information enters training, preprocessing, features, prompts, or inventory decisions.
Fair comparison	Same demand data, horizons, test periods, policy, cost assumptions, and metrics for every model.
Complexity earns its place	Models are added to the ladder only because they represent a methodological generation, not because they are fashionable.
No fabricated completeness	Unfinished analysis is marked unfinished. Invented values are never substituted.
Literature verified	Published claims are verified before use as evidence; not assumed true because they appear in a paper.

For the full research proposal: `01_research/research_proposal.md` For operational decisions and their rationale: `00_project/decisions.md`

Current progress

Phase	Status	Notebook(s)
0 — Repository reset	 Complete	RESET_LOG.md
Foundation — Environment	 Complete	01_environment_and_reproducibility.ipynb
Foundation — Data audit	 Complete	02_data_acquisition_and_audit.ipynb
Foundation — EDA	 Complete	03_exploratory_data_analysis.ipynb
Series selection	 Pending	04_series_selection_and_experimental_design.ipynb
Baseline — Naïve / Seasonal Naïve	 Pending	05_baseline_seasonal_naive.ipynb
Traditional — MA	 Pending	06_moving_average.ipynb
Traditional — SES	 Pending	07_simple_exponential_smoothing.ipynb
Traditional — DES	 Pending	08_double_exponential_smoothing.ipynb
Traditional — TES	 Pending	09_triple_exponential_smoothing.ipynb
Statistical — ARIMA	 Pending	10_arima.ipynb
Statistical — SARIMA	 Pending	11_sarima.ipynb
Classical comparison	 Pending	12_classical_models_comparison.ipynb
Neural — LSTM	 Pending	13_lstm.ipynb
LLM — Forecasting	 Pending	14_llm_forecasting.ipynb
Forecast comparison	 Pending	15_forecast_comparison.ipynb
Inventory simulation	 Pending	16_inventory_simulation.ipynb
Inventory results	 Pending	17_inventory_results.ipynb
Sensitivity analysis	 Pending	18_sensitivity_analysis.ipynb

Phase	Status	Notebook(s)
Statistical analysis	Pending	19_statistical_analysis.ipynb
Cross-dataset robustness	Pending	20_cross_dataset_robustness.ipynb
Final research findings	Pending	21_final_research_findings.ipynb

Legend:  Complete ·  In progress ·  Not started ·  Blocked

Key decisions so far

Date	Decision	Status
2026-08-22	Research question, model ladder, datasets, neutrality	Confirmed
2026-08-22	Conceptual freeze (proposal v0.1)	Confirmed
2026-08-22	Operational freeze v1.0 (h=28, n=50, L=7, α =95%, costs)	Reversed
2026-08-27	Clean research restart — notebook-first	Confirmed
2026-08-27	Cross-dataset comparability constraint (shared window, equal observations, evidence-driven sample size)	Confirmed

All experimental parameters (horizon, sample size, splits, costs, policy) are **open** and will be decided through EDA → literature → validation → feasibility → evidence, each recorded in `00_project/decisions.md` *before* use.

For the full decision log: `00_project/decisions.md`

Unresolved questions

1. **Dataset 02 feasibility** — 500 series, 1826 days, complete panel. Cross-dataset analysis with M5 feasible.
2. **Common window** — what is the optimal training window length given the dataset constraints?
3. **Series sample size** — how many series should be in the experimental sample? (Evidence-driven, not pre-fixed.)
4. **Rolling origin** — is rolling-origin evaluation computationally feasible given the dataset sizes and model complexity?

- 5. **Intermittent demand** — should a Croston/TSB-type model be added to the ladder given the high zero-demand share?
- 6. **LLM feasibility** — can a local LLM produce structured, valid numerical forecasts for this task?

References and documentation

Document	Location	Purpose
Research proposal (v0.1)	01_research/research_proposal.md	Conceptual source of truth
Experiment protocol (v1.0, superseded)	docs/research/experiment-protocol.md	Historical operational design
Decision log	00_project/decisions.md	All methodological decisions
Progress tracker	00_project/research_progress.md	Current status
Reset log	RESET_LOG.md	Record of the clean restart
Supporting code	11_src/	Config, metrics, plotting, data utilities

This notebook is the dashboard. It does not produce experimental results. Updated: 2026-08-27